CO1_AT1_Short Answer Test-Application based Question

1. A smartphone camera captures an image with very low spatial resolution, causing small objects to appear unclear. Apply the concept of sampling to identify the problem and suggest a suitable improvement.

2. A surveillance camera produces a dark and noisy image at night. Apply an appropriate image-processing technique to enhance the image and explain why it is suitable.

3. A medical image contains continuous intensity values that need to be converted into digital values. Apply the concept of quantization and explain how it helps in digital image formation.

4. An industrial camera captures a product with uneven lighting, making some defects difficult to see. Apply an image enhancement technique to improve defect visibility and state why it is suitable.

5. A self-driving car camera captures an image containing roads, vehicles, and pedestrians. Apply image segmentation to separate these objects and explain how it helps autonomous driving.

6. A factory inspection system needs to identify cracks on manufactured components. Apply edge detection to the captured image and explain how it can help detect the defects.

7. A medical image contains unwanted noise that makes it difficult to identify a tumor. Apply a suitable filtering technique and explain how it improves the image.

8. A traffic monitoring camera captures vehicles under different lighting conditions. Apply object detection to identify and track vehicles, and explain its importance in traffic management.

9. An image captured through a camera appears blurred because of object movement during acquisition. Identify the possible cause and suggest a suitable image-processing technique to improve the image.

10. An industrial vision system must distinguish defective products from good products based on their appearance. Apply image classification and explain how it can support automated quality inspection.